

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



Department of Information and Communication Technology

LAB Report

Lab Report No: 03

Lab Report Title: Multithreading Simulation: Car Parking Management System.

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

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Date of Performance:

Date of Submission:

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Lab Title : Multithreading - Car Parking Management System.

Objective :

To simulate a car parking system using multithreading.

Theory:

Multithreading allows multiple tasks to run concurrently. Threads share resources using synchronization. This helps simulate real-time systems.

Program Code:

```
class ParkingPool {
    synchronized void parkCar(String car, String agent) {
        System.out.println (agent + " parked car " + car);
    }
}

class ParkingAgent extends Thread {
    ParkingPool pool;
    String carName;

    ParkingAgent (ParkingPool p, String car) {
        pool = p;
        carName = car;
    }

    public void run() {
        pool.parkCar (carName, Thread.currentThread().getName());
    }
}
```



```

public class MainClass {
    public static void main (String[] args) {
        ParkingPool pool = new ParkingPool ();
        System.out.println ("Car ABC123 requested parking.");
        System.out.println ("Car XYZ456 requested parking.");

        ParkingAgent a1 = new ParkingAgent (pool, "ABC123");
        ParkingAgent a2 = new ParkingAgent (pool, "XYZ456");

        a1.setName ("Agent 1");
        a2.setName ("Agent 2");

        a1.start ();
        a2.start ();
    }
}

```

Output:

Car ABC123 requested parking.
 Car XYZ456 requested parking.
 Agent 1 parked car ABC123
 Agent 2 parked car XYZ456

Conclusion:

Multithreading helps simulate concurrent activities. Synchronization avoids conflicts while sharing resources.